

Hardware Requirements & Specifications (HRS)

Reaction Time (LED Chaser) Game — FPGA Digital Design Document

Project Name	Reaction Time (LED Chaser) Game
Target Architecture	FPGA (Compatible with Spartan-6 / Intel DE10-Standard)
Hardware Description Language	Verilog HDL
Designer	Abdelrahman (Boda)

1. System Overview

The Reaction Time Game is a digital hardware design aimed at testing human reflex speed. The system continuously shifts a single active LED across a 10-LED array at a fixed visible frequency. The user is challenged to press a push button at the exact moment the active LED aligns with a pre-configured target switch. The system evaluates the timing accuracy and outputs a result (Win/Lose) onto a multiplexed 4-digit 7-segment display.

2. Top-Level I/O Interfaces

Signal Name	Direction	Width	Description
CLK_50MHz	Input	1-bit	Main system clock (50 MHz).
RST_N	Input	1-bit	Active-low asynchronous reset. Resets the game state.
BTN_PRESS	Input	1-bit	Push button input for player reaction.
SW_TARGET	Input	10-bit	Sliding switches used to set the target LED position before the game starts.
LEDS	Output	10-bit	Output to 10 LEDs representing the shifting light.
SEG_DATA	Output	7-bit	Data lines for the 7-segment display (A–G).
SEG_SEL	Output	4-bit	Select/Enable lines for the 4 multiplexed 7-segment displays.

Block Diagram

Reaction Time (LED Chaser) Game — Top-Level Block Diagram

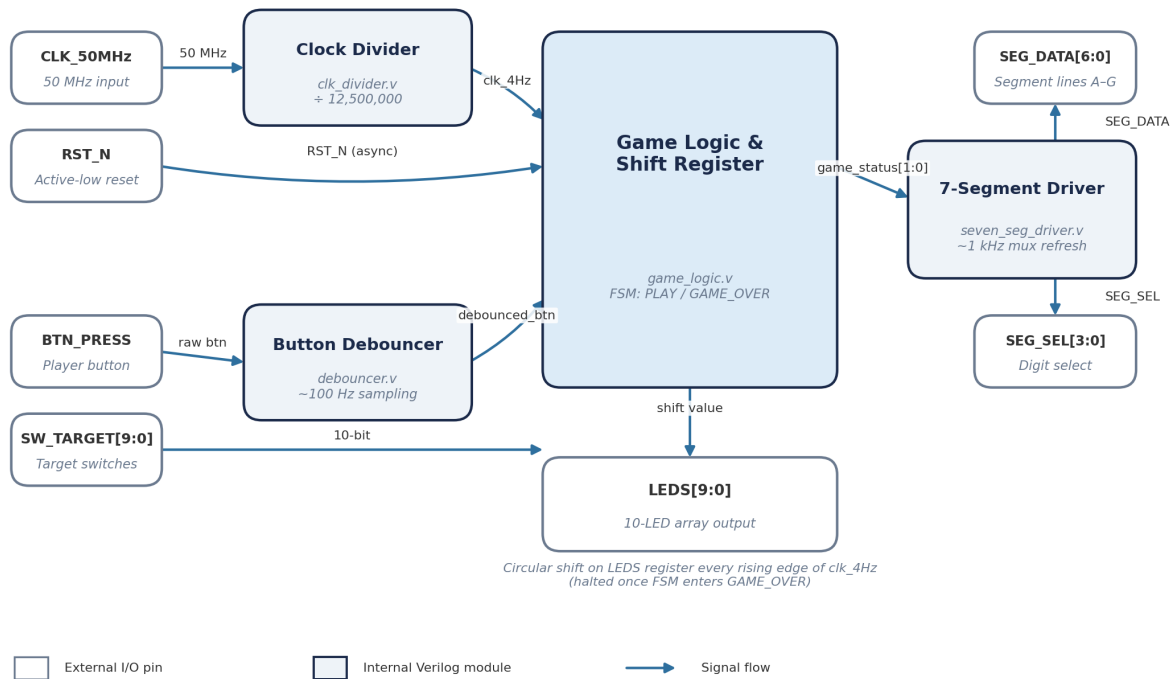


Figure 1. Top-level block diagram of the Reaction Time Game system.

3. Sub-Modules Specifications

3.1. Clock Divider (clk_divider.v)

- **Input Frequency:** 50 MHz (CLK_50MHz).
- **Target Frequency:** 4 Hz (Shift rate).
- **Division Factor:** 12,500,000.
- **Behavior:** Uses a synchronous counter. Toggles the output clock (clk_4Hz) every time the counter reaches $(12,500,000 / 2) - 1$ to maintain a 50% duty cycle.

3.2. Button Debouncer (debouncer.v)

- **Function:** Filters mechanical switch bounce from BTN_PRESS to prevent multiple false trigger events.
- **Sampling:** Uses a slower clock (e.g., 100 Hz) to sample the button state using a shift register (D-Flip Flops).
- **Output:** Generates a clean, single clock-cycle pulse (debounced_btn) when a stable press is detected.

3.3. Game Logic & Shift Register (game_logic.v)

- **FSM States:** PLAY, GAME_OVER.
- **Initial State:** On RST_N == 0, LEDS register is initialized to 10'b10_0000_0000 and FSM enters PLAY.
- **Shift Operation:** While in PLAY state, a circular right shift is performed on the LEDS register at every positive edge of clk_4Hz (e.g., LEDS <= {LEDS[0], LEDS[9:1]}).
- **Evaluation (Comparator):** Triggered asynchronously by debounced_btn while in PLAY state.

Condition	Result
LEDS == SW_TARGET	game_status is set to WIN
LEDS != SW_TARGET	game_status is set to LOSE

FSM transitions to GAME_OVER, halting the clk_4Hz shift operation.

3.4. Seven-Segment Display Driver (seven_seg_driver.v)

- **Function:** Drives the 4-digit 7-segment display using time-division multiplexing based on game_status.
- **Refresh Rate:** Requires a dedicated clock divider (approx. 1 kHz) to switch between SEG_SEL anodes/cathodes without human-visible flicker.

game_status	Display Output
WIN	B – O – S – S
LOSE	L – O – S – E
PLAY	Blanked / hyphens (– – – –)

4. Standard Operating Procedure (SOP)

- 1 **Target Setup:** The player configures SW_TARGET by setting exactly one switch to Logic 1 and the rest to Logic 0.
- 2 **System Initialization:** The player presses the reset button (RST_N = 0 then 1). The first LED illuminates, and the game enters the active state.
- 3 **Active Play:** The active LED shifts across the 10-bit array at 4 Hz.
- 4 **Reaction Event:** The player monitors the moving LED and presses BTN_PRESS when the illuminated LED aligns with the target switch.
- 5 **Resolution:** The logic freezes the LED position, evaluates the comparator output, and continuously displays the result on the 7-segment display until the next hardware reset.